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Pumping tests in non-uniform aquifers — the linear strip case

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ABSTRACT

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Many pumping tests are performed in geologic settings that can be conceptualized as a linear infinite strip of one material embedded in a matrix of differing flow properties. A semi-analytical solution is presented to aid the analysis of drawdown data obtained from pumping tests performed in settings that can be represented by such a conceptual model. Integral transform techniques are employed to obtain a solution in transform space that can be numerically inverted to real space. Examination of the numerically transformed solution reveals several interesting features of flow in this configuration. If the transmissivity of the strip is much higher than that of the matrix, linear and bilinear flow are the primary flow regimes during a pumping test. If the contrast between matrix and strip properties is not as extreme, then radial flow should be the primary flow mechanism. Sensitivity analysis is employed to develop insight into the controls on drawdown in this conceptual model and to demonstrate the importance of temporal and spatial placement of observations. Changes in drawdown are sensitive to the transmissivity of the strip for a limited time duration. After that time, only the total drawdown remains a function of strip transmissivity. In the case of storativity, both the total drawdown and changes in drawdown are sensitive to the storativity of the strip for a time of quite limited duration. After that time, essentially no information can be gained about the storage properties of the strip from drawdown data. An example analysis is performed using data previously presented in the literature to demonstrate the viability of the semi-analytical solution and to illustrate a general procedure for analysis of drawdown data in complex geologic settings. This example reinforces the importance of observation well placement and the time of data collection in constraining parameter correlation, a major source of the uncertainty that arises in the parameter estimation procedure.

INTRODUCTION

The pumping test is the primary tool used by hydrogeologists to characterize the large-scale transmissive and storage properties of a hydrologic unit. The techniques used to analyze the drawdown data collected during and after a period of pumping are based on analytical solutions to partial differential equations describing the flow of ground water to a central pumping well. A major assumption incorporated into these analytical solutions is that the

transmissive and storage properties are uniform in the unit of interest, i.e. the properties are invariant in space. Although this assumption has been shown to be viable in many situations, the geological complexity of some units may make this assumption of dubious value. The purpose of this article is to explore methods for the analysis of pumping-test data in a class of geological settings that are not amenable to the conventional analytical approach.

This article examines pumping tests in geologic settings that can be conceptualized as an infinite linear strip of one material embedded in a matrix of differing hydraulic properties. This conceptual model can be considered to represent the major features of several different geologic settings such as a buried stream channel embedded in a less-permeable matrix, an igneous dike intruded into a country rock of differing properties, and a pair of parallel communicating faults separating a central strip of material from a dissimilar matrix. If the conceptual model is altered slightly such that the linear strip is adjoined by different materials on either side, this model can also be considered to represent large-scale trends in material properties. As this conceptual model represents a wide class of geologic settings, it is important that techniques are developed for the analysis of response data from a pumping test in this configuration.

The research reported in this article is built on a large body of previous work in the hydrogeology and petroleum engineering literature. A number of workers (e.g. Maximov, 1962; Bixel et al., 1963; Nind, 1965; Fenske, 1984) have examined the effects of pumping in a configuration in which a single linear discontinuity separates materials of dissimilar properties. Yaxley (1987) and Ambastha et al. (1989) have modified this model somewhat by adding a positive skin at the discontinuity to represent a partially communicating boundary. A considerable number of workers (e.g. Bochever, 1959; Ferris et al., 1962; Chan et al., 1976; Tiab and Kumar, 1980; Streltsova and McKinley, 1984; Ehlig-Economides and Economides, 1985) have also examined the case of pumping from a strip of permeable material bounded on at least two parallel sides by impermeable boundaries. Two relatively recent contributions have touched upon pumping in the specific configuration that is the focus of this article. Clarke (1987) proposed an image well approach for assessing the effects of pumping in a strip aquifer embedded in a matrix of differing properties. However, he offered no rigorous mathematical proof for his approach, so it must be considered only an approximate representation of the actual behavior. Boonstra and Boehmer (1986) presented an approximate solution for a similar configuration in which the transmissivity of the strip must be much larger than that of the matrix. Apparently, only limited work has been done to assess the effects of pumping in a unit with large-scale property trends. Hantush (1962), in one of the few articles touching on this

issue, presented an approximate solution to the case of pumping in an aquifer of exponentially increasing thickness.

This article presents a general semi-analytical solution for the conceptual model consisting of a central infinite strip bounded on either side by semi-infinite units of differing properties. There are no limitations on the magnitude of the property contrast between the different materials or on the dimensionality of the flow in the three units. As the solution presented here is based on a straightforward extension of the integral transform approach of Bixel et al. (1963), among others, the details of the solution are relegated to the Appendix. There are two major objectives of this article. The first is to discuss the insight into flow behavior in the important class of geologic settings represented by this conceptual model that can be gained from the solution presented here. The second objective is to propose a general approach for the analysis of pumping-test data in this configuration.

STATEMENT OF THE PROBLEM

The problem of interest here is that of the drawdown, as a function of x , y , and t , produced by pumping from a fully penetrating well in the configuration displayed in Fig. 1. The pumping well can be located in any of the three zones illustrated in the figure. Flow properties are assumed to be uniform within each zone, although differing between zones. The variation in flow properties between zones can be of any magnitude. Apart from the variation in flow properties across zonal boundaries, the standard conditions for a pumping test in a confined aquifer are assumed at all times. Flow within the zones that do not contain the pumping well can be described mathematically

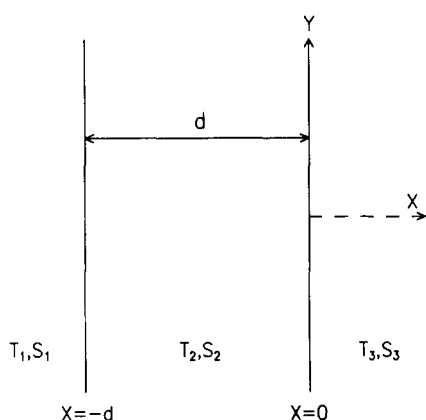


Fig. 1. Configuration consisting of an infinite strip of material 2 (width = d) separating two semi-infinite half-spaces of materials 1 and 3. Note that the y axis lies on the boundary between materials 2 and 3.

by the standard Cartesian coordinate form of the aquifer flow equation:

$$\frac{\partial^2 s_i}{\partial x^2} + \frac{\partial^2 s_i}{\partial y^2} = \frac{S_i \partial s_i}{T_i \partial t} \quad (1)$$

where s_i is the drawdown in material i , [L]; t is the time, [T]; S_i is the storage coefficient of material i , dimensionless; T_i is the transmissivity of material i , [L² T⁻¹]. Flow within the zone that contains the pumping well is represented in the manner employed by Bixel et al. (1963) and Chan et al. (1976). As shown in eqn. (2), a source term with Dirac delta functions is used to represent the pumping well rather than a boundary as is the standard procedure in the well hydraulics literature:

$$\frac{\partial^2 s_i}{\partial x^2} + \frac{\partial^2 s_i}{\partial y^2} + \frac{Q}{T_i} \delta(x - a) \delta(y - b) = \frac{S_i \partial s_i}{T_i \partial t} \quad (2)$$

where Q is the pumpage from well located at (a, b) , [L³ T⁻¹]; $\delta(x - a) = 0$ when $x \neq a$ or $\delta(x - a) = 1$ when $x = a$, [L⁻¹]; $\delta(y - a) = 0$ when $y \neq a$ or $\delta(y - a) = 1$ when $y = a$, [L⁻¹].

The initial conditions are the same for all three zones and can be written as

$$s_i(x, y, 0) = 0 \quad (3)$$

The boundary conditions, which vary from zone to zone, can be written as

$$s_1(-\infty, y, t) = 0 \quad (4)$$

$$s_i(x, \pm \infty, t) = 0 \quad (5)$$

$$s_3(\infty, y, t) = 0 \quad (6)$$

$$s_1(-d, y, t) = s_2(-d, y, t) \quad (7)$$

$$s_2(0, y, t) = s_3(0, y, t) \quad (8)$$

$$T_1 \frac{\partial s_1(-d, y, t)}{\partial x} = T_2 \frac{\partial s_2(-d, y, t)}{\partial x} \quad (9)$$

$$T_2 \frac{\partial s_2(0, y, t)}{\partial x} = T_3 \frac{\partial s_3(0, y, t)}{\partial x} \quad (10)$$

Note that two conditions must be met at zonal boundaries to ensure continuity of flow between the zones. Also note that, as shown in Fig. 1, the central strip is assumed to extend from $-d$ to 0 in the x direction. There are no limitations on the size of the central strip.

Equations (1)–(10) describe the flow conditions of interest here. The Appendix provides the details of the solution approach applied in this work. In summary, the solution approach employs a series of integral transforms (Laplace transform in time and Fourier transform in the y direction), in a manner similar to that of Bixel et al. (1963), to obtain functions in transform space that satisfy the transform-space analogues of (1)–(10). The Fourier–Laplace space functions that are obtained for the case of a pumping well located in the central strip can be written in a non-dimensional form as

$$\overline{\phi}_1 = \frac{D\rho_1}{\lambda_2} \left[\frac{(1 - \rho_2)e^{-\lambda_2\alpha} - (1 + \rho_2)e^{\lambda_2\alpha}}{\text{Denom}} \right] e^{-\lambda_2 + \lambda_1 + \lambda_1\zeta} \quad (11)$$

$$\begin{aligned} \overline{\phi}_2 = \frac{D}{2\lambda_2(\text{Denom})} \{ & (1 + \rho_1)[(1 - \rho_2)e^{-\lambda_2(\alpha - \zeta)} - (1 + \rho_2)e^{-\lambda_2|\alpha + \zeta|}] \\ & - (1 - \rho_1)[(1 - \rho_2)e^{\lambda_2|\zeta + \alpha|} - (1 + \rho_2)e^{\lambda_2(\alpha - \zeta)}] e^{-2\lambda_2} \} \end{aligned} \quad (12)$$

$$\overline{\phi}_3 = \frac{-D\rho_2}{\lambda_2} \left[\frac{(1 + \rho_1)e^{-\lambda_2\alpha} - (1 - \rho_1)e^{-2\lambda_2 + \lambda_2\alpha}}{\text{Denom}} \right] e^{-\lambda_3\zeta} \quad (13)$$

where $\overline{\phi}_i$ is the Fourier–Laplace transform of ϕ_i , the non-dimensional form of s_i ; $D = (-e^{i\omega\beta})/(p\sqrt{2\pi})$; p is the Laplace transform variable; ω is the Fourier transform variable; (α, β) are the dimensionless (x, y) coordinates of the pumping well; $\text{Denom} = (1 + \rho_1)(1 + \rho_2) - (1 - \rho_1)(1 - \rho_2)e^{-2\lambda_2}$; $\lambda_1 = (\omega^2 + P_1p)^{0.5}$; $\lambda_2 = (\omega^2 + p)^{0.5}$; $\lambda_3 = (\omega^2 + P_2p)^{0.5}$; $P_1 = \mu_1/\mu_2$; $P_2 = \mu_3/\mu_2$; $\gamma_i = T_{i+1}/T_i$, $i = 1, 2$; $[\rho_i = (\gamma_i\lambda_{i+1})/\lambda_i]$, $i = 1, 2$; $\mu_i = S_i/T_i$, $i = 1, 2, 3$.

For equations of the complexity of (11)–(13), the analytical back transformation from Fourier–Laplace space to real space is only readily performed under quite limited conditions. In the general case, the transformation is best performed numerically. For this work, the Fourier inversion was performed using Romberg integration (Carnahan et al., 1969). An algorithm developed by Stehfest (1970), which has been found to be of great use in hydrologic applications (Moench and Ogata, 1984), was employed to perform the numerical Laplace inversion.

As discussed in the Introduction, Boonstra and Boehmer (1986) and Clarke (1987) have presented approximate solutions to flow in a configuration similar to that shown in Fig. 1. To assess the impact of the approximations incorporated in their solutions, a comparison between their solutions and the form derived here was performed. The upper pair of curves in Fig. 2 illustrates the results of a comparison between the solution of Boonstra and Boehmer (1986) and that of this work for a four-order-of-magnitude contrast in transmissivity

between the center strip and the matrix. The lower pair of curves in Fig. 2 illustrates a similar comparison for a transmissivity contrast of two orders of magnitude. Note that the Boonstra and Boehmer solution becomes more approximate as the contrast decreases. This is to be expected as their solution was designed for conditions of a high transmissivity strip embedded in a much less transmissive matrix. For small to moderate dimensionless times and for large transmissivity contrasts, the solution of Boonstra and Boehmer seems to be a reasonable approximation of system behavior.

Figure 3 illustrates a comparison between the solution of Clarke (1987) and that of this work for a configuration similar to that used for the lower pair of curves in Fig. 2. Note that the two solutions differ considerably at large times. The magnitude of the difference is a function of the property contrast across the zonal boundaries. Unlike the previously discussed solution, the solution of Clarke converges on the actual solution as the magnitude of the property contrast across the zonal boundaries decreases. Although the solution of Clarke is parallel to the actual solution at large times, the discrepancy in the drawdown predictions makes his solution of limited value.

The above comparisons indicate that neither of the previously presented solutions can be employed as a general tool for examining flow conditions in the conceptual model displayed in Fig. 1. Therefore, the solution derived here is used for the remainder of this paper to examine several issues concerning flow in this configuration.

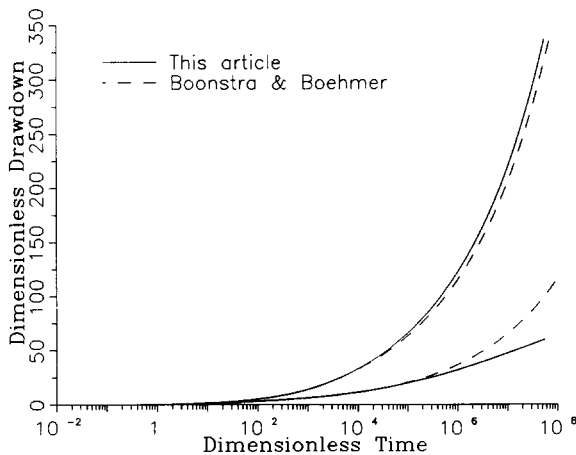


Fig. 2. Dimensionless drawdown $[(4\pi s_2 T_2)/Q]$ versus time $(4T_2 t/S_2 d^2)$ plots comparing the solution presented in this article with that given by Boonstra and Boehmer (1986) ($S_1 = \text{constant}$, $x/d = -0.43$, $y/d = 0.5$, $a/d = -0.5$, $b/d = 0$). The upper pair of curves is the comparison for case of strip transmissivity being four orders of magnitude larger than that of the matrix; the lower pair is the comparison for case of strip transmissivity being two orders of magnitude larger than that of the matrix.

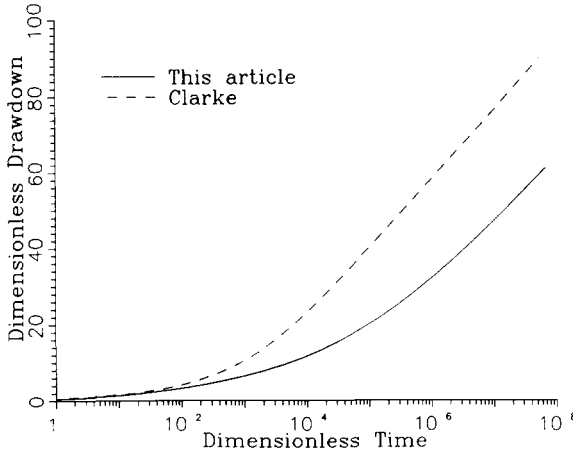


Fig. 3. Dimensionless drawdown $[(4\pi s_2 T_2)/Q]$ versus time $(4T_2 t/S_2 d^2)$ plot comparing the solution presented in this article with that given by Clarke (1987) for the case of strip transmissivity being two orders of magnitude larger than that of the matrix ($S_i = \text{constant}$, $x/d = -0.43$, $y/d = 0$, $a/d = -0.5$, $b/d = 0$).

DISCUSSION OF THEORETICAL RESPONSES

Figure 4 serves as a good starting point for the discussion of several issues concerning flow in the configuration shown in Fig. 1. The specific configuration examined in Fig. 4 is illustrated in the inset. Note that, except for curve 2, the

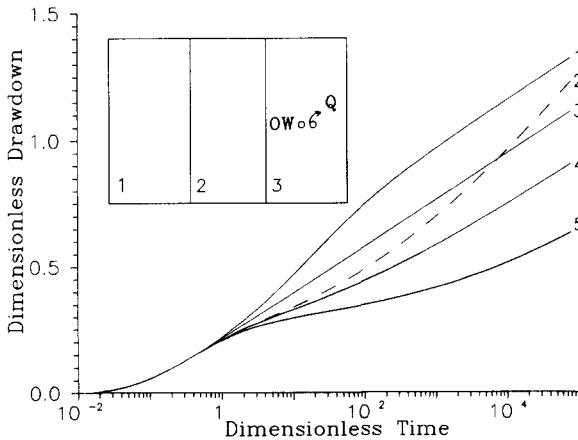


Fig. 4. Dimensionless drawdown $[(4\pi s_3 T_3)/Q]$ versus time $(4T_3 t/S_3 d^2)$ plots for the case of a pumping well (denoted by Q) at a dimensionless location $(1.4, 0)$ and an observation well (denoted by OW) at a dimensionless location $(1, 0)$ ($S_i = \text{constant}$, curve 1 — $T_2 = 0.1T_1 = 0.1T_3$; curve 2 — $T_2 = 10T_1 = 100T_3$; curve 3 — $T_i = \text{constant}$ (uniform aquifer); curve 4 — $T_2 = 10T_1 = 10T_3$; curve 5 — $T_2 = 100T_1 = 100T_3$).

matrix properties remain constant on either side of the central strip. The curves in Fig. 4 all display the same general features: a relatively early-time straight line on the semi-log plot, a transition zone that may or may not display a straight-line segment, and then a large-time straight line that may be parallel to the first straight line. The first straight-line segment reflects the infinite-acting radial flow that occurs before the front of the cone of depression (the term used here to denote the frontal portion of the pressure disturbance created by pumping from which the majority of flow to the pumping well originates) passes through the right-hand boundary of the central strip. A Cooper–Jacob semi-log analysis (Cooper and Jacob, 1946) of this segment would yield the properties of the matrix. The transition zone that separates the small and large time linear regions actually consists of three segments: an initial segment reflecting the passage of the front of the cone of depression across the right-hand boundary, a second segment when the system is behaving as if it consists of two units separated by a single linear discontinuity, and a final segment reflecting the passage of the front of the cone of depression across the left-hand boundary. The nature of drawdown during the first and last segments of this transition period is a function of the difference in transmissivity and storativity between the matrix and the strip. Whether or not a linear segment is clearly displayed during this transition period is a function of the width of the central strip and the property contrasts between the strip and matrix. If a straight line is shown, a Cooper–Jacob analysis will yield a transmissivity that is an arithmetic average of the strip and matrix (Streltsova, 1988).

The third general flow regime displayed in Fig. 4 is revealed by the large-time straight-line segment. Except for curve 2, the large-time straight-line segment is parallel to the first straight-line segment, indicating that a Cooper–Jacob analysis at large times will yield a transmissivity equal to that of the matrix. As changes in drawdown are used in the Cooper–Jacob analysis, this result demonstrates that changes in drawdown at large times are independent of strip properties. Later, it is shown that the total drawdown at large times is dependent on strip transmissivity but independent of strip storativity. The contribution of the strip to total drawdown is reflected in the offset between the early-time and large-time straight-line segments. If the properties of the matrix differ on either side of the strip, the transmissivity calculated by a Cooper–Jacob analysis will be an arithmetic average of the transmissivities in the two semi-infinite half spaces. This is illustrated by curve 2 where the slope of the large-time straight line is a function of the arithmetic average of the transmissivities in the left and right semi-infinite half spaces. Note that the analysis of the theoretical responses displayed in Fig. 4 indicates that the semi-log method first proposed by Cooper and Jacob (1946) for uniform aquifers

is applicable to a wide variety of non-uniform systems if the flow to the pumping well is approximately radial during the period of the analysis. Although this point has long been recognized in the petroleum literature (e.g. Streltsova, 1988, and references therein), it has received limited discussion in the groundwater literature (e.g. Butler, 1988). Also note that the arithmetic average discussed in the preceding paragraphs is an unweighted average whose form is not dependent on the location of pumping and observation wells.

The behavior displayed in Fig. 4 is rather ideal as a straight-line segment can be discerned for at least two of the three general flow regimes. The actual drawdown behavior observed at an observation well depends on the position of the pumping and observation wells, the magnitude of the contrast between the strip and the matrix, and the direction of the contrast (i.e. the strip transmissivity is larger than that of the matrix and vice versa). Contour maps of pumping-induced drawdown can help illustrate how drawdown is dependent on well position for different contrasts between matrix and strip properties.

Figure 5 displays the large-time drawdown for the case when the transmissivity of the central strip is four orders of magnitude greater than that of the matrix. Notice how the flow in both the strip and the matrix has a strong linear component, a feature that is the basis of the approximate solution of Boonstra and Boehmer (1986). If the transmissivity contrast is large, then a log-log drawdown plot may display a straight-line segment with a slope of 0.5, as shown in Fig. 6. A straight line with a slope of 0.5 on a log-log plot is a signature of linear flow (Gringarten, 1985). The half-slope straight line in Fig. 6, however, is different from the very-early-time straight line of slope 0.5 that is often associated with a high conductivity vertical fracture intersecting the well bore. Instead, the half-slope straight line observed in Fig. 6 is analogous to the moderate- to large-time straight line discussed by Ehlig-Economides and Economides (1985). Note that the half-slope line in Fig. 6 is succeeded by a quarter-slope line, which is diagnostic of bilinear flow (Boonstra and Boehmer, 1986). The bilinear flow regime exists when the flow is approximately linear in both the strip and the matrix, as in Fig. 5. This moderate- to large-time quarter-slope line is different from the very-early-time quarter-slope straight line that is associated with a low-conductivity vertical fracture intersecting the well bore (Gringarten, 1985). The duration of the linear and bilinear flow regimes is a function of the magnitude of the contrast between strip and matrix properties. If an observation well is located within the strip at a large distance from the pumping well, the linear flow regime will be difficult to observe unless the transmissivity contrast is extremely large. Therefore, in distant observation wells, the appearance of the bilinear flow regime will be the primary indication of a large contrast between strip and matrix transmissivity.

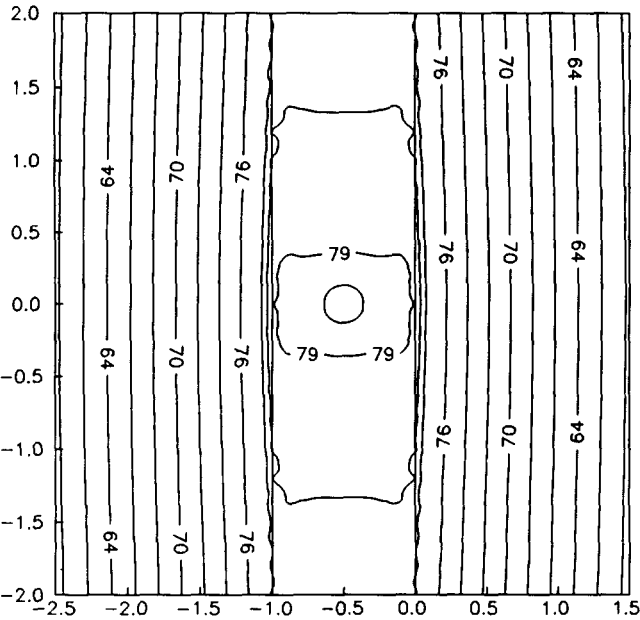


Fig. 5. Contour map of dimensionless drawdown $[(4\pi s_2 T_2)/Q]$ after a large duration of pumping (dimensionless time $(4T_2 t/S_2 d^2) = 2e + 7$) for the case of the strip transmissivity much larger than that of the matrix ($T_2 = 10000T_1 = 10000T_3$, $S_i = \text{constant}$, distances normalized by d , pumping well assumed at $(-0.5, 0)$, varying contour interval (2 from 60 to 78, 0.5 from 78 to 79, and 0.2 from 79 to 79.2) used to emphasize structure of pumping-induced drawdown field). Note that the small oscillations especially prominent at the corners of the closed contours are an artifact of the contouring routine.

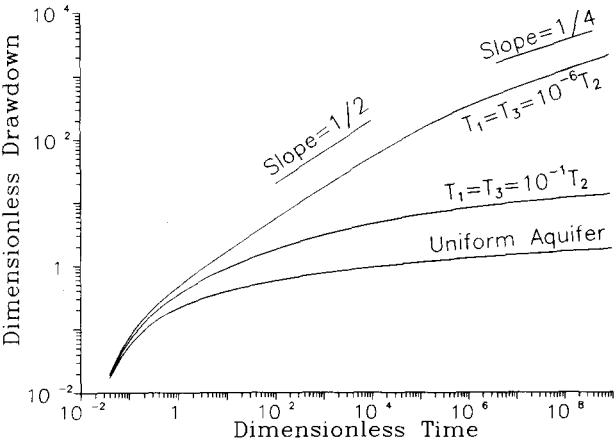


Fig. 6. Dimensionless drawdown $[(4\pi s_2 T_2)/Q]$ versus time $(4T_2 t/S_2 d^2)$ plot displaying the linear and bilinear flow regimes for the case of the strip transmissivity much larger than that of the matrix (pumping well at $(-0.5, 0)$, observation well at $(-0.1, 0)$; reference lines used to emphasize linear and bilinear flow regimes).

Figure 7 displays the drawdown for the case when the transmissivity of the central strip is two orders of magnitude greater than that of the matrix. Although there is considerable refraction of the head contours across the strip boundaries, the pumping-induced flow has a much stronger radial component than in the previous case. There is still, however, a strong linear component to the flow owing to the fact that the strip is serving as a major conduit for the system. If the matrix transmissivity is greater than that of the strip, as in Fig. 8, the head contours take on a considerably different look, reflecting the fact that the majority of the flow is now occurring in the matrix and not in the strip. Note that in the configuration shown in Fig. 8, water originating within the strip near the pumping well is flowing directly to the well. At greater distances from the well, however, water originating in the strip flows into the more transmissive matrix before returning to the strip in the vicinity of the pumping well. The distance from the pumping well at which the more circuitous flow path becomes the most efficient depends primarily upon the magnitude of the transmissivity contrast and the strip width. Note also that, unlike the configurations displayed in Figs. 5 and 7, the solution for the case shown in Fig. 8 will converge on the uniform aquifer case as the width of the less transmissive strip is reduced.

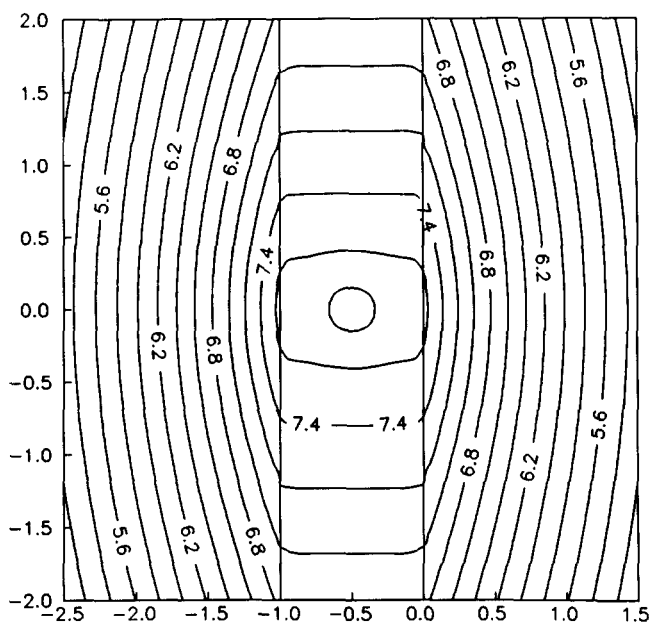


Fig. 7. Contour map of dimensionless drawdown $[(4\pi s_2 T_2)/Q]$ after a moderate duration of pumping (dimensionless time $(4T_2 t/S_2 d^2) = 200$) for the case of the strip transmissivity two orders of magnitude larger than that of the matrix ($T_2 = 100T_1 = 100T_3$, $S_1 = \text{constant}$, distances normalized by d , pumping well assumed at $(-0.5, 0)$, contour interval of 0.2).

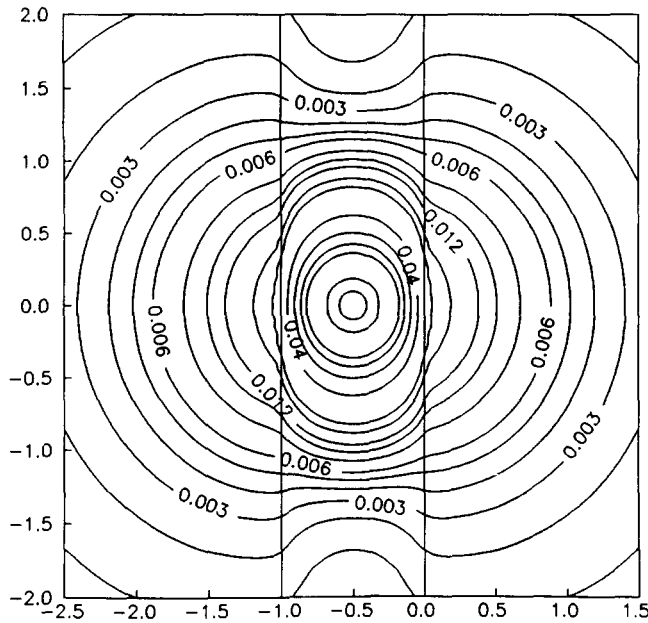


Fig. 8. Contour map of dimensionless drawdown $[(4\pi s_2 T_2)/Q]$ after a moderate duration of pumping (dimensionless time $(4T_2 t/S_2 d^2) = 0.2$) for the case of the strip transmissivity one order of magnitude smaller than that of the matrix ($T_2 = 0.1T_1 = 0.1T_3$, $S_i = \text{constant}$, distances normalized by d , pumping well assumed at $(-0.5, 0)$, varying contour interval (0.001 from 0.001 to 0.006, 0.002 from 0.006 to 0.012, 0.004 from 0.012 to 0.02, 0.02 from 0.02 to 0.1, and 0.1 from 0.1 to 0.3) used to emphasize structure of pumping-induced drawdown field).

SENSITIVITY ANALYSIS OF THEORETICAL RESPONSES

Further insight into flow in this configuration can be obtained by applying the principles of sensitivity analysis. Sensitivity analysis (McElwee and Yukler, 1978; McElwee, 1987) is a formalism that allows the relationship between drawdown and flow properties to be examined in considerable detail. The sensitivity of drawdown at (x, y, t) to changes in a given parameter such as the transmissivity of zone 1 ($\partial s(x, y, t)/\partial T_1$) is termed a sensitivity coefficient. The multiplication of a sensitivity coefficient by the parameter of interest is defined as a normalized sensitivity coefficient (McElwee, 1987). The normalized sensitivity coefficient is utilized in the discussions of this work to examine the dependence of drawdown on flow properties in the different zones of the conceptual model.

Figure 9 depicts the results of an analysis of the sensitivity of drawdown at an observation well to the properties of the central strip. Curves a–d of Fig. 9 illustrate the sensitivity of drawdown to the transmissivity of the central strip for different observation and pumping well configurations, and for different

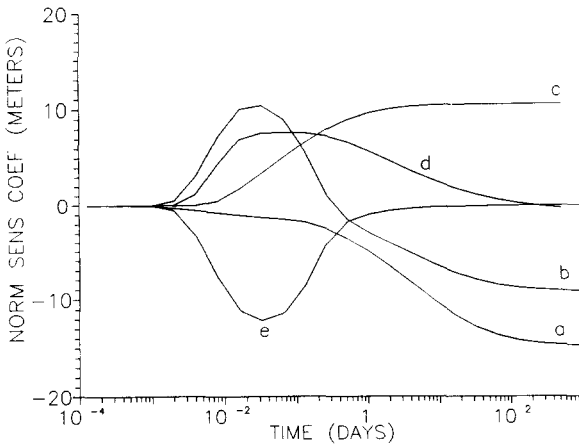


Fig. 9. Sensitivity to strip properties versus time plot. Curves a–d depict sensitivity to strip transmissivity and curve e depicts sensitivity to strip storativity ($S_s = 5e - 4$, $T_1 = T_3 = 10 \text{ m}^2 \text{ day}^{-1} = 0.1T_2$, $d = 10 \text{ m}$, $Q = 1000 \text{ m}^3 \text{ day}^{-1}$; curve a, pumping well at (5, 0), observation well at (2, 0); curve b, pumping well at (5, 0), observation well at (-2, 0); curve c, pumping well at (5, 0), observation well at (-15, 0), $T_1 = T_3 = 10 \text{ m}^2 \text{ day}^{-1} = 10T_2$; curve d, pumping well at (-5, 0), observation well at (-15, 0), $T_1 = T_3 = 10 \text{ m}^2 \text{ day}^{-1} = 10T_2$; curve e, pumping well at (5, 0), observation well at (-2, 0)).

contrasts between the transmissivity of the strip and the matrix. For an observation and pumping well in the same half space (curve a), the transmissivity of the strip will influence changes in drawdown for a finite period of time. The flattening of the sensitivity curve indicates that further changes in drawdown will be independent of T_2 , although the total drawdown will always be dependent on T_2 . This sensitivity behavior is analogous to that seen in radial flow systems for zones lying radially outward from the pumping and observation wells (Butler and McElwee, 1990). The duration of the period in which changes in drawdown are significantly influenced by the transmissivity of the strip is a function of strip width and the contrast between the properties of the strip and those of the matrix. If the observation well is located in the strip itself, the general response will be similar to that given by curve b. At early times, drawdown at the observation well will have a positive sensitivity whereas at later times it will change to negative and flatten, as in curve a. The positive sensitivity is produced as the front of the cone of depression passes the observation well. After the front of the cone of depression has passed the observation well, the sensitivity relationships are negative. If the observation well is in the half space opposite that of the well, two sensitivity relationships can be seen. For the case of the transmissivity of the strip being considerably larger than that of the matrix, the sensitivity relationships will be analogous to those seen in curve a. For the case of T_2 lower than that of the matrix (curve c), the sensitivities will always be positive.

In the cases described in the preceding paragraph, the pumping well is in the matrix. When both the pumping and observation wells are in the central strip, the sensitivity plot is analogous to that seen in curve b of Fig. 9, although the magnitude of the sensitivity will be greater. When the observation well is in the matrix and the pumping well is in the strip, two types of sensitivity relationships can be seen. For the case of the strip being more conductive than the matrix, the sensitivity relationships are analogous to those depicted in curve a. The total drawdown at the observation well will always be dependent on T_2 . For the case of the strip being less conductive than the matrix, the sensitivity relationships are quite different (curve d). In this case, T_2 influences both changes in drawdown and total drawdown at the observation well for a time of limited duration. After that time, the drawdown at the observation well is essentially independent of T_2 . This independence at large times arises because the majority of the water being pumped is originating within the matrix, and the flow within the matrix is essentially radial towards the pumping well at all points (see Fig. 8).

Curve e of Fig. 9 is a general depiction of the sensitivity of drawdown to the storativity of the central strip. In all the pumping and observation well permutations discussed above, there is only a finite duration of time during which changes in drawdown and total drawdown are dependent on S_2 . After that time, no information about the storage properties of the central strip can be gained by drawdown measurements. Although the amplitude and width of the curve and the timing of the peak will vary from case to case, the general relationship is that shown by curve e of Fig. 9.

Sensitivity analysis can also be used to examine the relationship between drawdown and various geometrical aspects of the flow system, such as the strip width. Figure 10 displays three curves that summarize the sensitivity of drawdown to strip width for different observation and pumping well configurations, and for different contrasts between the transmissivity of the strip and the matrix. In all cases, there is only a small interval of time during which changes in drawdown are sensitive to strip width. After that interval, further changes in drawdown are independent of strip width. A comparison of the magnitude of the normalized sensitivity coefficients of Figs. 9 and 10 indicates that under most conditions drawdown is significantly less sensitive to strip width than to strip transmissivity.

The preceding discussion has focused on the sensitivity of drawdown to the properties and width of the central strip. The sensitivity of drawdown to the properties of the two half spaces bounding the central strip is similar to that found in an infinite uniform aquifer. In this case, the sensitivity to transmissivity will not level off but will continue to increase throughout the period of the test (McElwee and Yukler, 1978). The sensitivity to storativity, on the other hand, will level off to a constant value after a relatively small time.

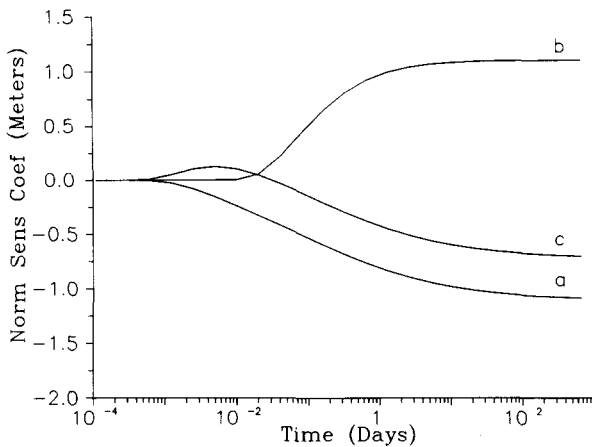


Fig. 10. Sensitivity to strip width versus time plot ($S_i = 5e - 4$, $d = 10$ m, $Q = 1000 \text{ m}^3 \text{ day}^{-1}$; curve a, pumping well at (5, 0), observation well at (2, 0), $T_1 = T_3 = 10 \text{ m}^2 \text{ day}^{-1} = 0.1T_2$; curve b, pumping well at (5, 0), observation well at (2, 0), $T_1 = T_3 = 10 \text{ m}^2 \text{ day}^{-1} = 10T_2$; curve c, pumping well at (5, 0), observation well at (-15, 0), $T_1 = T_3 = 10 \text{ m}^2 \text{ day}^{-1} = 0.1T_2$).

The sensitivity analysis of the preceding paragraphs can provide some insight into the spatial and temporal placement of observations in a geologic setting similar to that depicted in Fig. 1. In all cases when the properties of the central strip are of interest, it is clear that attention must be given to collecting data during the interval of time when changes in drawdown are sensitive to strip properties (i.e. when the sensitivity coefficients are changing). As Butler and McElwee (1990) have shown for radial systems, collection of drawdown data outside this interval of changing sensitivities will be of limited usefulness. If the pumping well is located in the central strip, most information about the entire system will be obtained from an observation point within the strip near the pumping well. Observation wells in the matrix or in the strip at a considerable longitudinal distance from the pumping well will provide less information about strip properties. If the pumping well is located in the matrix, observation wells are best placed near the pumping well in the same half space or within the portion of the strip closest to the pumping well. If the properties of the matrix are the subject of interest, an observation well close to the pumping well is recommended to obtain data that are most sensitive to matrix properties.

DATA ANALYSIS

For cases such as that shown in Fig. 4, where several semi-log straight lines are clearly displayed on the drawdown versus time plots, the transmissivity of the different units controlling drawdown response can be readily estimated. However, in those cases where the pumping well is placed near a

boundary, when the transitions between the different flow regimes is long, or when information other than just the transmissivities of the various units is needed, a more sophisticated analysis must be performed. This section outlines an approach for drawdown analysis based on the automated matching of theoretical type curves to the drawdown data.

For the solution presented here to be of much use for estimating hydraulic parameters, some geologic information must be known about the system that is being tested. If the width of the strip is known along with the position of the pumping and observation wells with respect to the strip, then this solution may provide useful information about the flow system. A computer program is available (Liu and Butler, 1990) that will allow the user to generate theoretical responses that can be fitted to the drawdown data. If this program is implemented with an automated fitting routine, much of the tedious nature of the curve matching can be avoided. Owing to the computationally intensive nature of the solution, an automated matching procedure cannot be performed in a single sitting on a microcomputer. However, the matching can be completed by allowing a PC-AT or more powerful microcomputer to run for several hours. The speed of the matching procedure can be increased significantly if some of the parameters can be estimated by an initial semi-log analysis, as described earlier. Those parameters can then be considered as known quantities in the matching procedure. If none of the parameters can be estimated before the automated matching procedure, large correlation in the sensitivity of drawdown to several of the parameters may result in very large error bounds being associated with the estimates of those parameters. Butler and McElwee (1990) discussed the effects of sensitivity magnitude and sensitivity correlation on estimates of hydraulic parameters from pumping tests.

If a conceptual model similar to that of Fig. 1 is suspected to be an appropriate representation of the flow system, but little additional information is known, it may be difficult to use automated curve matching as an exploratory tool to provide information concerning the position and width of the strip, as well as the relevant flow parameters. The non-uniqueness of the drawdown behavior and the high correlation between the various parameters would greatly hinder efforts of this type. In this case, the best alternative is to employ a variety of drawdown data plots in different formats in an attempt to identify general flow regimes and initial parameter estimates. If high-quality data have been obtained, drawdown-derivative plots (Clark and Van Golf-Racht, 1985; Bourdet et al., 1989) should be used. These plots of the temporal derivative of drawdown versus time may display features (which are not as evident on the conventional plots) that enable the flow system to be better defined. Several examples of drawdown-derivative plots are given here to illustrate their use as an exploratory and data analysis tool.

Figure 11 displays both drawdown and drawdown-derivative log-log plots for the case of pumping from a high-permeability channel embedded in a lower-permeability matrix. Note that the drawdown-derivative plot has several distinctive features that provide some clues to the nature of the flow regime during particular intervals of time. The interval denoted A on Fig. 11 marks the period of time during which the primary flow regime is radial flow to the pumping well. The horizontal line on the drawdown-derivative plot is a characteristic feature of radial flow and is easier to recognize than a straight line on a semi-log plot. As discussed by Bourdet et al. (1989), transmissivity can be estimated from the vertical position of the horizontal line. The abrupt ending of the horizontal line is an indication that some sort of boundary is controlling system behavior. The interval denoted B marks a period of time during which the primary flow regime is linear flow to the pumping well. The half-slope line on the drawdown-derivative plot is diagnostic of linear flow and may be easier to recognize than the half-slope on the drawdown plot discussed earlier. The gradual decrease of the drawdown-derivative curve at the end of interval B is an indication that flow from the matrix is beginning to contribute to the discharge. The interval denoted C marks a period of time during which the primary flow regime is bilinear. The quarter-slope line on the drawdown-derivative plot is diagnostic of bilinear flow and again may be easier to recognize than its analogue on the drawdown plot. If the transmissivity contrast between the strip and matrix is considerably less, the intervals denoted B and C only last for a short period of time and are succeeded by a transition to another horizontal line, indicating that radial flow within the

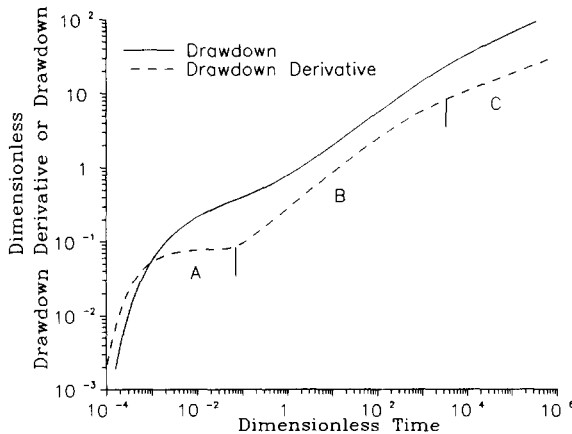


Fig. 11. Dimensionless drawdown $[(4\pi s_2 T_2)/Q]$ and derivative of drawdown $[(4\pi T_2/Q)(\partial s_2/\partial t)]$ versus time $(4T_2 t/S_2 d^2)$ plot for the case of the strip transmissivity much larger than that of the matrix ($T_2 = 10000T_1$, $S_1 = 10000T_3$, $S_1 = \text{constant}$, pumping well assumed at $(-0.5, 0)$, observation well at $(-0.46, 0)$).

matrix is the primary flow mechanism. The vertical position of the second horizontal line is a function of the transmissivity of the matrix. Note that although the features mentioned above can all be identified using other means by a knowledgeable observer, the features are often more readily identified on the drawdown-derivative plots.

Figure 12 displays both drawdown and drawdown-derivative log-log plots for the case of pumping from a high-permeability matrix in the vicinity of a lower-permeability strip. Note that again the drawdown-derivative plot has several distinctive features that provide some clues to the nature of the flow regime during particular intervals of time. In this case, however, the features are more subtle. The interval denoted A on Fig. 12 marks the initial period of radial flow before the effects of the lower permeability strip are felt. The increase in slope at the end of interval A indicates that a boundary with a less-permeable unit has been reached. The interval denoted B marks the period of time during which the lower-permeability unit has a significant influence on changes in drawdown. The marked decrease at the end of this interval indicates that a boundary with a more permeable unit has been reached. The subsequent horizontal line shows that radial flow through the matrix is again the primary flow mechanism. The similar vertical position of the two horizontal lines is an indication that the transmissivities of the two half-spaces comprising the matrix are similar.

As illustrated in the preceding discussion, drawdown-derivative plots can provide important additional information about the flow regime during certain intervals of the pumping period. Given that some of the diagnostic

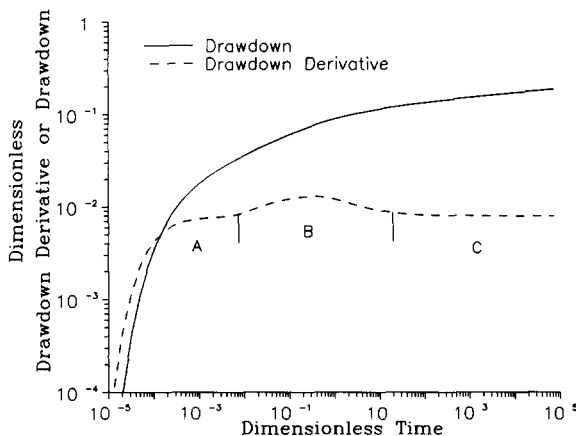


Fig. 12. Dimensionless drawdown $((4\pi s_3 T_3)/Q)$ and derivative of drawdown $[(4\pi T_3/Q)(\partial s_3/\partial t)]$ versus time $(4T_3 t/S_3 d^2)$ plot for the case of the strip transmissivity smaller than that of the matrix ($T_2 = 0.1T_1 = 0.1T_3$, $S_i = \text{constant}$, pumping well assumed at $(0.45, 0)$, observation well at $(0.40, 0)$).

features may be rather subtle and that noise in the measurements will be accentuated by the derivative calculation, it is imperative that drawdown measurements be taken with higher accuracy and higher frequency than is normally done in groundwater applications. Bourdet et al. (1989) discussed several techniques for minimizing the effects of noise on the drawdown-derivative calculations. If high quality data have been obtained, not only can the drawdown-derivative plots be used to identify flow regimes and to provide some initial estimates of system parameters, but also can be used in an automated curve-matching procedure similar to that done with conventional log-log drawdown curves. If the parameters resulting from the drawdown and drawdown-derivative matches are not in close agreement, this should be taken as evidence that the chosen model may be inappropriate and that further analysis is required.

EXAMPLE APPLICATION

In this section, a data set available in the literature is employed to demonstrate the viability of the solution presented here and to illustrate a general strategy for drawdown analysis using the conceptual model of Fig. 1.

Boonstra and Boehmer (1986) presented drawdown data from a pumping test in the Brandwag Tweeling dike in South Africa. The pumped well was located in a dolerite dike 10 m wide embedded in a lower permeability matrix of sandstones and mudstones. Although data from both the pumping well and an observation well in the dike, 100 m from the pumping well, were presented, only the observation well data are employed here owing to uncertainty concerning possible well losses at the pumping well. Table 1 lists the drawdown data from the observation well and the discharge rate.

The first step in a drawdown analysis is that of model selection. In this case, given the geologic data that are available, it is apparent that the conceptual model of Fig. 1 (with zones 1 and 3 having the same properties) is the appropriate model. As this model requires that four parameters be determined from the drawdown data, it is recommended that various plots of the drawdown data be studied before the automated curve matching. As discussed previously, these data plots may enable preliminary estimates of one or more of the parameters to be made. In Fig. 13(a), a semi-log plot of observation well drawdown versus time is shown. No straight-line segments are found that would enable estimation of strip or matrix transmissivity. Figure 13(b) displays drawdown and drawdown-derivative log-log plots. Note that the log-log drawdown plot does appear to indicate that bilinear flow is occurring for a portion of the test, thus supporting the choice of the conceptual model of Fig. 1 as the appropriate model. For this test, the

TABLE 1

Drawdown data from Brandwag Tweeling dike pumping test

Time (min)	Drawdown (m)	Time (min)	Drawdown (m)
1	1.3780	150	9.0630
2	2.0680	175	9.5530
3	2.5070	200	10.045
4	2.8180	250	11.027
6	3.3600	300	11.627
8	3.8460	350	12.154
10	4.2240	400	12.207
13	4.5470	500	14.324
15	4.7650	600	15.031
18	5.0160	750	15.907
21	5.2570	900	15.704
25	5.5190	1050	17.813
30	5.7000	1200	17.565
35	6.0440	1350	17.916
40	6.2320	1500	17.945
50	6.6060	1700	18.285
60	6.9070	1900	18.409
75	7.3490	2100	18.483
100	8.0310	2300	18.858
125	8.8850	2500	19.109

$Q = 1200 \text{ m}^3 \text{ day}^{-1}$; both pumping and observation well in the dike; dike width = 10 m.

drawdown-derivative plot yields little information owing to the noise in the drawdown data. Note that a variety of smoothing algorithms suggested by Bourdet et al. (1989) were employed to calculate the temporal derivative of drawdown. However, the drawdown data were too sparse and too noisy to be of use, even after processing using the smoothing algorithms.

The automated matching of theoretical type curves to the drawdown data is the next step in a drawdown analysis. In this case, as no parameter estimates can be obtained from the various drawdown plots, all four parameters must be estimated in the matching procedure. The curve-matching procedure employed here uses an iterative non-linear least-squares approach to obtain the best match of a theoretical type curve to the field data (Bohling et al., 1990). An initial value for each of the parameters must be assigned before the first iteration. To improve the speed of the convergence and to prevent convergence on local minima, it is helpful to provide initial values that incorporate any information that exists about the flow system. In this test, a drawdown of 1.38 m was measured at the observation well 1 min after the

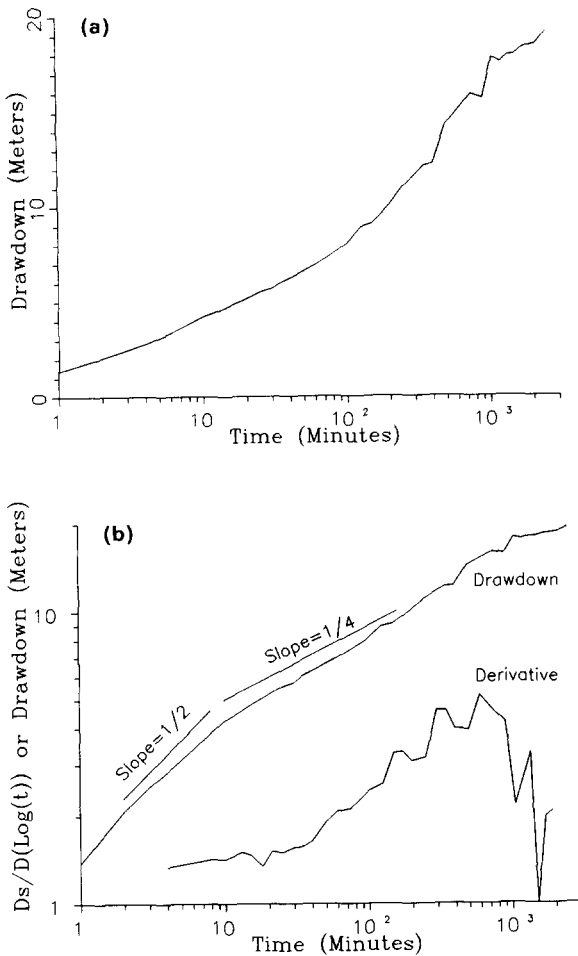


Fig. 13. Data plots for Brandwag Tweeling pumping test: (a) semi-log drawdown versus time plot; (b) log-log drawdown and derivative of drawdown versus time plot (reference lines indicate slope of linear ($1/2$) and bilinear ($1/4$) flow regimes).

initiation of pumpage. This large rapid drawdown indicates that the strip has a high transmissivity and a very low storage. The apparent bilinear flow at moderate to large times indicates that the matrix transmissivity is considerably less than that of the strip. Table 2 lists the initial parameter values that were employed for this analysis. Using these initial values, the matching program converged on the set of parameters given in Table 2. Figure 14 displays a comparison of the drawdown data and the theoretical curve obtained from the matching procedure. Except at large times when drawdown begins to level off, the visual match is quite good. Note that, except for the estimate of strip storativity, the values determined here are similar to those

TABLE 2
Details of analysis of Brandwag Tweeling pumping test data

	Parameter			
	$T_1 = T_3$ (m ² day ⁻¹)	T_2 (m ² day ⁻¹)	$S_1 = S_3$	S_2
Initial value	0.10e - 2	1	0.50e - 3	0.50e - 4
Final value ^a	0.70e - 2	1.4	0.54e - 4	0.25e - 6
Lower bound ^b	-0.24e - 2	-1.25	-0.15e - 4	-0.24e - 3
Upper bound ^b	0.16e - 1	4.06	0.12e - 3	0.24e - 3

^a Root-mean-squared deviation of 0.55 m.
^b Lower and upper bounds represent 95% confidence interval.

determined by Boonstra and Boehmer (1986) with their approximate model, although those authors were unable to estimate the properties of the matrix beyond the product of storativity and transmissivity. Unfortunately, as shown in Table 2, the approximate confidence intervals about the obtained estimates are extremely broad, even extending into the range of physical implausibility. As discussed earlier, the width of the confidence intervals is strongly influenced by the correlation between the sensitivity of drawdown to the various parameters. As correlation appears to be greatly hindering efforts to estimate hydraulic parameters reliably in this test, it is of interest to plot the sensitivity coefficients for this test in order to examine this correlation in more detail.

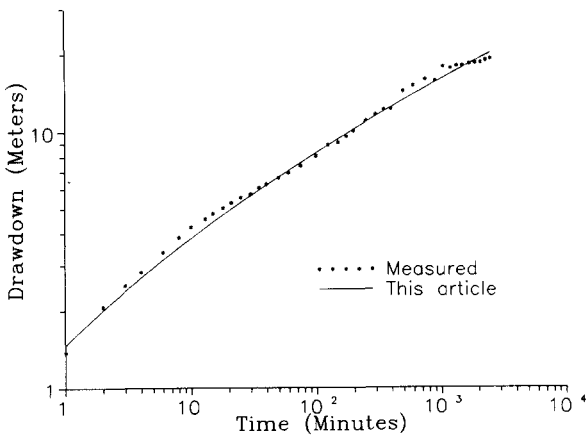


Fig. 14. Log-log drawdown versus time plot comparing the measured drawdown with the theoretical drawdown calculated using the parameters determined from the non-linear least-squares fitting procedure.

Figure 15 is a plot of the sensitivity of drawdown to all four parameters determined in the analysis. The two vertical lines indicate the start and end of the period in which data were collected during the test at Brandwag Tweeling. The first point to note about this plot is that in the initial portions of the data collection period, the sensitivities to the transmissivity and storativity of the matrix behave in an almost identical fashion. Although the sensitivities begin to diverge at larger times, the continuing high correlation makes it difficult to isolate the effects of the individual parameters using the observation well that was available. The second point to note is that the sensitivity to the storativity in the strip is extremely low. The third point to note is that the sensitivity to the strip transmissivity is also highly correlated with the matrix properties over the period of measurement. It is only at early times that there is a significant difference in the sensitivity behavior between the matrix properties and the strip transmissivity. The high correlation between the matrix properties and the strip transmissivity in conjunction with the low sensitivity to the strip storativity clearly led to the large approximate confidence intervals shown in Table 2.

Figure 15 shows that unless data can be collected during a period of relatively low correlation, it may be almost impossible to estimate needed parameters reliably. This figure also indicates the importance of observation well placement in pumping tests. In this case, an observation well within the matrix is required in order to estimate the transmissive and storage properties of the matrix reliably. Even an observation well in the matrix, however, would be of little use unless drawdown data are collected over a time range that will minimize the correlation between matrix parameters.

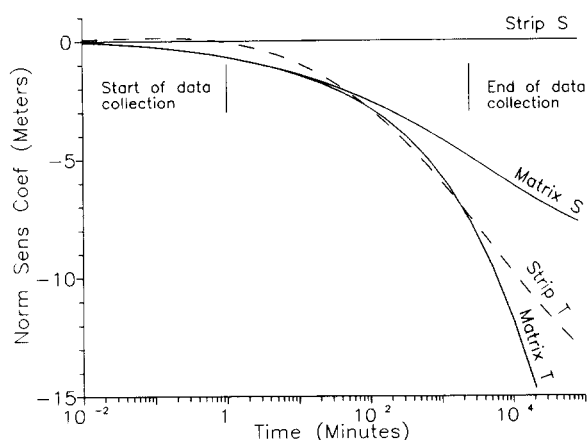


Fig. 15. Normalized sensitivity versus time plot for the four parameters estimated from the Brandwag Tweeling pumping-test data.

The sensitivity plots of Fig. 15 demonstrate the importance of pumping-test design for the determination of hydraulic parameters. The generation of sensitivity plots before a test using the best available estimates of system parameters may greatly aid in the design of a test that provides the most information about the parameters of interest. Without such a design phase, it may be difficult to assess a suitable spatial and temporal placement of observations.

SUMMARY AND CONCLUSIONS

A semi-analytical solution has been presented to aid in the analysis of drawdown data collected in geologic settings that can be conceptualized as an infinite linear strip of one material embedded in a matrix of differing hydraulic properties. This solution, which is based on an application of standard integral transform techniques, can increase understanding of flow in complex geologic systems. Linear and bilinear flow regimes will predominate in cases where the strip transmissivity is considerably larger than that of the matrix and the pumping well is located in the strip. For systems in which the transmissivity contrast is not as extreme, radial flow will be the primary flow regime. Each of these flow regimes has a diagnostic 'signature' that can be recognized using various types of drawdown plot. Sensitivity analysis can help identify appropriate temporal and spatial strategies for data collection in complex geologic systems. Data collection during the period of changing sensitivities with respect to the parameter of interest is required to obtain parameter estimates of high reliability. Changes in drawdown are only sensitive to strip properties for a time of limited duration, so care must be taken to acquire drawdown data during that interval. Even with such a data collection strategy, however, correlation between parameters can frustrate parameter estimation efforts. An approach for the analysis of drawdown data based on automated curve matching is recommended. The success of the automated curve-matching procedure is greatly enhanced if various data plots can provide estimates of some of the needed parameters. Drawdown data from a test previously reported in the literature help illustrate many of the issues discussed here. The results of this test clearly demonstrate the importance of a pre-test design phase for pumping tests in complex geologic media. Sensitivity analysis should be a crucial element of such design work.

Although the conceptual model examined here incorporates important features from a variety of geologic settings, it cannot include all possibilities. In many systems, the central strip has a finite length, which may influence drawdown at moderate to large durations of pumpage. When the solution given here is considered in conjunction with an earlier study of the impact on

drawdown of a circular pod embedded in a matrix of differing properties (Butler and Liu, 1989), it is clear that the response in the case of a strip of finite length will not be dramatically different from the solution given here. The major difference will be that the period in which the strip properties influence changes in drawdown will decrease. A feature of many fluvial systems is the interconnection of channel sands over a large area. If the transmissivity of the sands does not change dramatically along the axis of the channel deposits, the observed behavior should be similar to that examined here. Highly anastomosed channel deposits, however, may be better represented by other models.

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APPENDIX

In this section, the mathematical derivation of the solution discussed in this article is presented. For the sake of generality, the solution is obtained in a dimensionless form. After the derivation of a solution in Fourier–Laplace space has been presented, the reduction of the solution to simpler forms, which may be analytically inverted to real space, is discussed. For the general

case, however, the inversion is best done numerically. The Appendix concludes with a brief discussion of the numerical inversion technique employed here.

In this derivation, a solution is presented for the case of a pumping well located anywhere within the central strip of Fig. 1. Equations (1)–(10) describe the flow conditions of interest here. To work with the most general form of the solution, this derivation is performed using dimensionless forms of (1)–(10). The dimensionless analogues of (1)–(10) are as follows:

$$\frac{\partial^2 \phi_1}{\partial \xi^2} + \frac{\partial^2 \phi_1}{\partial \eta^2} = P_1 \frac{\partial \phi_1}{\partial \tau} \quad (\text{A1.1})$$

$$\frac{\partial^2 \phi_3}{\partial \xi^2} + \frac{\partial^2 \phi_3}{\partial \eta^2} = P_2 \frac{\partial \phi_3}{\partial \tau} \quad (\text{A1.2})$$

$$\frac{\partial^2 \phi_2}{\partial \xi^2} + \frac{\partial^2 \phi_2}{\partial \eta^2} + \delta(\xi - \alpha)\delta(\eta - \beta) = \frac{\partial \phi_2}{\partial \tau} \quad (\text{A2})$$

$$\phi_i(\xi, \eta, 0) = 0 \quad (\text{A3})$$

$$\phi_1(-\infty, \eta, \tau) = 0 \quad (\text{A4})$$

$$\phi_i(\xi, \pm\infty, \tau) = 0 \quad (\text{A5})$$

$$\phi_3(\infty, \eta, \tau) = 0 \quad (\text{A6})$$

$$\phi_1(-1, \eta, \tau) = \phi_2(-1, \eta, \tau) \quad (\text{A7})$$

$$\phi_2(0, \eta, \tau) = \phi_3(0, \eta, \tau) \quad (\text{A8})$$

$$\frac{\partial \phi_1(-1, \eta, \tau)}{\partial \xi} = \gamma_1 \frac{\partial \phi_2(-1, \eta, \tau)}{\partial \xi} \quad (\text{A9})$$

$$\frac{\partial \phi_2(0, \eta, \tau)}{\partial \xi} = \gamma_2 \frac{\partial \phi_3(0, \eta, \tau)}{\partial \xi} \quad (\text{A10})$$

where

$$P_1 = \frac{\mu_1}{\mu_2}$$

$$P_2 = \frac{\mu_3}{\mu_2}$$

$$\gamma_i = \frac{T_{i+1}}{T_i}$$

$$\tau = \frac{tT_2}{d^2 S_2}$$

$$\phi_i = \frac{s_i T_2}{Q}$$

$$(\xi, \eta) = \left(\frac{x}{d}, \frac{y}{d} \right)$$

$$(\alpha, \beta) = \left(\frac{a}{d}, \frac{b}{d} \right)$$

$$\mu_i = \frac{S_i}{T_i}, \quad i = 1, 2, 3$$

Note that the central strip has a width of unity in the dimensionless system.

A solution can be obtained for (A1)–(A10) through the use of integral transforms (Robinson, 1968). A Laplace transform in time followed by an exponential Fourier transform in the y direction produces Fourier–Laplace space analogues to (A1) and (A2) of the following form:

$$\frac{\partial^2 \overline{\phi}_i}{\partial \xi^2} - \lambda_i^2 \overline{\phi}_i = 0, \quad i = 1 \text{ and } 3 \quad (\text{A11})$$

$$\frac{\partial^2 \overline{\phi}_2}{\partial \xi^2} - \lambda_2^2 \overline{\phi}_2 = - \frac{e^{i\omega\beta} \delta(\xi - \alpha)}{p\sqrt{2\pi}} \quad (\text{A12})$$

where $\lambda_1 = (\omega^2 + P_1 p)^{0.5}$; $\lambda_2 = (\omega^2 + p)^{0.5}$; $\lambda_3 = (\omega^2 + P_2 p)^{0.5}$; ω is the Fourier-transform variable and p is the Laplace-transform variable.

The Fourier–Laplace space solution to (A11) for both zones 1 and 3 is quite straightforward. After incorporating the boundary conditions of (A4) and (A6), the expressions for zones 1 and 3 can be written as follows:

$$\overline{\phi}_1 = C_1 e^{\lambda_1 \xi}, \quad \xi \leq -1 \quad (\text{A13})$$

$$\overline{\phi}_3 = C_7 e^{-\lambda_3 \xi}, \quad \xi \geq 0 \quad (\text{A14})$$

The Fourier–Laplace space solution to (A12), however, cannot be found as easily owing to the non-homogeneous delta function term in that expression. The approach used here for obtaining a solution to (A12) was to divide the central strip into two subregions, subregion 21 ($-1 \leq \xi \leq -\alpha$) and subregion 22 ($-\alpha < \xi \leq 0$). The solution for subregion 21 consists of a homogeneous part and a particular solution ($\overline{\phi}_{2p}$):

$$\overline{\phi}_{21} = C_3 e^{\lambda_2 \xi} + C_4 e^{-\lambda_2 \xi} + \overline{\phi}_{2p} \quad (\text{A15})$$

The particular solution can be written as

$$\overline{\phi}_{2p} = \begin{cases} \frac{-e^{i\omega\beta}}{2\lambda_2 p \sqrt{2\pi}} [e^{\lambda_2(\xi+\alpha)} - e^{-\lambda_2(\xi+\alpha)}], & \xi = -\alpha \\ 0, & \text{elsewhere} \end{cases} \quad (\text{A16})$$

The solution for subregion 22 consists solely of a homogeneous part and can be written as

$$\overline{\phi}_{22} = C_5 e^{\lambda_2 \xi} + C_6 e^{-\lambda_2 \xi} \quad (\text{A17})$$

The division of the central strip into two subregions requires that two additional boundary conditions be introduced to ensure continuity across the division in the central strip:

$$\overline{\phi}_{21}(-\alpha, \omega, p) = \overline{\phi}_{22}(-\alpha, \omega, p) \quad (\text{A18})$$

$$\frac{\partial \overline{\phi}_{21}(-\alpha, \omega, p)}{\partial \xi} = \frac{\partial \overline{\phi}_{22}(-\alpha, \omega, p)}{\partial \xi} \quad (\text{A19})$$

The constants in (A13)–(A15) and (A17) can be evaluated by substituting these expressions into the Fourier–Laplace space equivalents to (A7)–(A10):

$$\overline{\phi}_1(-1, \omega, p) = \overline{\phi}_{21}(-1, \omega, p) \quad (\text{A20})$$

$$\overline{\phi}_{22}(0, \omega, p) = \overline{\phi}_3(0, \omega, p) \quad (\text{A21})$$

$$\frac{\partial \overline{\phi}_1(-1, \omega, p)}{\partial \xi} = \gamma_1 \frac{\partial \overline{\phi}_{21}(-1, \omega, p)}{\partial \xi} \quad (\text{A22})$$

$$\frac{\partial \overline{\phi}_{22}(0, \omega, p)}{\partial \xi} = \gamma_2 \frac{\partial \overline{\phi}_3(0, \omega, p)}{\partial \xi} \quad (\text{A23})$$

and (A18) and (A19). Although a rather lengthy procedure, the evaluation of the constants is simply an algebraic exercise and therefore is not described here. Once the constants are found, they are substituted back into (A13)–(A15) and (A17) to produce the following expressions:

$$\overline{\phi}_1 = \frac{D\rho_1}{\lambda_2} \left[\frac{(1 - \rho_2)e^{-\lambda_2 \alpha} - (1 + \rho_2)e^{\lambda_2 \alpha}}{\text{Denom}} \right] e^{-\lambda_2 + \lambda_1 + \lambda_1 \xi} \quad (\text{A24})$$

$$\begin{aligned} \overline{\phi}_{21} = \frac{D}{2\lambda_2(\text{Denom})} \{ & (1 + \rho_1)[(1 - \rho_2)e^{-\lambda_2(\alpha - \xi)} - (1 + \rho_2)e^{\lambda_2(\alpha + \xi)}] \\ & - (1 - \rho_1)[(1 - \rho_2)e^{-\lambda_2(\xi + \alpha)} - (1 + \rho_2)e^{\lambda_2(\alpha - \xi)}] e^{-2\lambda_2} \} \end{aligned} \quad (\text{A25.1})$$

$$\overline{\phi}_{22} = \frac{D}{2\lambda_2(\text{Denom})} \{ (1 + \rho_1)[(1 - \rho_2)e^{-\lambda_2(x-\xi)} - (1 + \rho_2)e^{-\lambda_2(x+\xi)}] \\ - (1 - \rho_1)[(1 - \rho_2)e^{\lambda_2(\xi+x)} - (1 + \rho_2)e^{-\lambda_2(\xi-x)}]e^{-2\lambda_2} \} \quad (\text{A25.2})$$

$$\overline{\phi}_3 = \frac{-D\rho_2}{\lambda_2} \left[\frac{(1 + \rho_1)e^{-\lambda_2 x} - (1 - \rho_1)e^{-2\lambda_2 + \lambda_2 x}}{\text{Denom}} \right] e^{-\lambda_3 \xi} \quad (\text{A26})$$

where

$$D = \frac{-e^{i\omega\beta}}{p\sqrt{2\pi}}$$

$$\text{Denom} = (1 + \rho_1)(1 + \rho_2) - (1 - \rho_1)(1 - \rho_2)e^{-2\lambda_2}$$

$$\rho_i = \frac{\gamma_i \lambda_{i+1}}{\lambda_i}, \quad i = 1, 2$$

Equations (A25.1) and (A25.2) can now be combined into a single expression:

$$\overline{\phi}_2 = \frac{D}{2\lambda_2(\text{Denom})} \{ (1 + \rho_1)[(1 - \rho_2)e^{-\lambda_2(x-\xi)} - (1 + \rho_2)e^{-\lambda_2|x+\xi|}] \\ - (1 - \rho_1)[(1 - \rho_2)e^{\lambda_2|\xi+x|} - (1 + \rho_2)e^{\lambda_2(x-\xi)}]e^{-2\lambda_2} \} \quad (\text{A27})$$

Equations (A24), (A26), and (A27) therefore form the Fourier–Laplace space solutions to (A1)–(A10). Substitution of (A24), (A26), and (A27) back into the transform-space analogues of (A1)–(A10) demonstrates the viability of the solution given here. Note that these expressions were derived for the case of a pumping well in the central strip as this will probably be the most common application of the solution. Similar expressions, however, have been found for the case of the pumping well in one of the semi-infinite half-spaces. The solution for these cases is available from the authors on request.

To be of practical use, the Fourier–Laplace space solution must be inverted back to real space. For expressions of the complexity of those given here, the analytical back transformation is not readily performed for the general case. In this work, a numerical inversion scheme is employed for the general case. Before describing that approach, however, the restricted conditions for which (A24), (A26), and (A27) can be analytically back transformed are briefly discussed.

If the properties of all three zones are assumed to be the same, then it can be shown that (A24), (A26), and (A27) reduce to expressions that can be analytically back transformed to the solution of Theis (1935). If two adjacent zones have the same properties and these properties differ from those of the

remaining zone, the Fourier–Laplace solution given here reduces to eqns. (15) and (16) of Bixel et al. (1963), which those authors state can be analytically back transformed for the constant-diffusivity case. If the diffusivity (μ_i) of all three zones is set equal, then an analytical back transformation is readily performed when there is either a very large contrast between strip and matrix transmissivities or quite a small contrast.

The numerical inversion scheme used here consists of two steps: a Romberg integration procedure (Carnahan et al., 1969) to integrate the Fourier inversion integral numerically, and an implementation of the quadrature algorithm of Stehfest (1970) for the integral arising in the Laplace inversion. Liu and Butler (1990) have written a Fortran program that uses this procedure to invert expressions (A24), (A26), and (A27) numerically, as well as the analogous expressions that arise for the case of a pumping well in one of the semi-infinite half-spaces. This program has been checked against several of the cases described in the previous paragraph when an analytical back transformation is possible. The close agreement between the numerical and analytical inversion schemes for all cases that were examined is considered strong support for the viability of the numerical approach. Note that in this work, a relative convergence criterion of 10^{-9} was employed for the Romberg integration and 12 terms were used in the summation in the Stehfest algorithm. A larger number of terms in the Stehfest algorithm would have required a more stringent convergence criterion, as the Stehfest weighting factors will become quite large and tend to magnify small convergence errors. A useful rule of thumb is to keep the relative convergence criterion at least an order of magnitude smaller than the inverse of the largest Stehfest weighting factor. Note that increasing the Stehfest terms and the stringency of the convergence criterion will significantly increase the total time of the computation. For the purposes of this work, the very small gains in accuracy that can be achieved through this procedure were not considered worth the increased computational burden.